

Problem 7.1

At an effective rate of interest of 8% per annum, the present value of \$100,000 due in X years is \$65,322. Determine X .

Solution

Using the present value relationship,

$$100000 = 65322(1.08)^X.$$

Taking natural logarithms on both sides,

$$\ln\left(\frac{100000}{65322}\right) = X \ln(1.08).$$

Therefore,

$$X = \frac{\ln\left(\frac{100000}{65322}\right)}{\ln(1.08)}.$$

Evaluating,

$$X \approx 5.5332.$$

Hence,

$$\boxed{X \approx 5.5332 \text{ years}}.$$